# StudentInsight

# Student Performance Prediction

*Technical report covering five milestone chapters*

Bindura University of Science Education

Programme  Network Engineering

Module  Artificial Intelligence and Intelligent System

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Submission  Group project

Group number _____

Group members and student numbers

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This report follows the supplied M1 to M5 structure. It documents the implemented system and measured results. Complete group identities and role allocations before submission.

# Abstract and report guide

StudentInsight is a staff-facing student-performance analysis system for teachers, lecturers, heads of department, principals and headmasters. It estimates final grades after the second grading period and combines individual forecasts with class tables, charts, reference-grade alerts and downloadable reports. Python, pandas, NumPy, scikit-learn, Streamlit, Altair and SQLite support the implementation. Jupyter and Google Colab reuse the same model code, with optional Google Drive file access.

The study evaluates 395 Mathematics and 649 Portuguese subject records. Six procedures share five-fold validation on the fitting partition: training mean, G2 persistence, two ridge regressions and two random forests. Selection uses validation mean absolute error and a predefined simplicity rule. G2 persistence wins for both subjects. Test MAE is 1.000 for Mathematics and 0.677 for Portuguese. At nominal 80% interval coverage, observed coverage is 83.5% and 93.8%, respectively, with intervals of constant width two grade points.

These are internal benchmarks on previously inspected test records. They do not establish reliability for Bindura University, local schools or future cohorts. The contribution is a working analysis workflow whose input requirements, predictions, uncertainty statements and exports agree with its evidence. All 56 available automated tests passed on 26 October 2026. Google-hosted authorization and the assessed live presentation remain activities to complete in the relevant user sessions.

| Chapter | Milestone | Content |
|---|---|---|
| 1 | M1 Proposal  Week 4 | Problem, objectives, methodology, roles and literature |
| 2 | M2 Design  Week 7 | Architecture, pipeline, algorithms and ethics |
| 3 | M3 Prototype  Week 10 | Implementation, training and evaluation |
| 4 | M4 Integration  Week 13 | Integration, tests, documentation and repository |
| 5 | M5 Final  Week 15 | Final package, conclusions, slides and demo |

The week numbers are planned assessment points supplied in the brief. They do not assert that historical submissions occurred on those dates. References follow the five chapters and count within the technical report limit.

# Chapter 1 Proposal

### 1 1 Background and problem statement

Educators review grades alongside information about learning conditions. StudentInsight addresses the need for a consistent final-grade estimate and a clear account of its supporting evidence. It is intended to organize academic review conversations, rather than automate promotion, discipline or admission. No institutional needs survey or participant interviews have been completed; this is a defined project use case rather than a measured description of university practice.

The earlier prototype selected learned regressors but evaluated the G2 comparator only after selection. A more complex model could therefore be deployed even when repeating G2 offered a better reference forecast. The project also needed consistent feature requirements, separate class-alert evaluation and a clear distinction between recorded grades, predictions and intervals. These concrete modelling and software problems motivated the completed repair.

### 1 2 Aim and objectives

The aim is to develop and evaluate an understandable student-performance prediction and analysis system for educational staff, using information available after period two.

- Validate recorded grades and supported contextual variables without manufacturing missing data.
- Compare mean, persistence, ridge and forest candidates on identical training folds, selecting by validation MAE.
- Measure errors and interval coverage with separate fitting, calibration and test partitions.
- Provide individual and class analyses with charts, reference alerts and consistent exports.
- Integrate authentication, versioned artifacts, notebooks and optional Drive storage, then verify meaningful user flows.
- Document limitations and the genuine data needed before institutional use.

Research questions ask whether learned candidates improve validation error over strong baselines, which inputs the selected predictor requires, what test error and interval coverage it achieves, and whether staff can inspect and export consistent results. Chapter 3 addresses the modelling questions and Chapter 4 documents functional verification.

# Proposal scope and methodology

### 1 3 Scope and significance

Mathematics and Portuguese are independently modelled public subjects. English, Physics, Chemistry, Biology, Geography, History and Computer Science require completed subject history before predictions become available. Mathematics is never relabelled as another subject. Grades are whole numbers from zero to twenty, and recorded G2 is mandatory for the current prediction timing.

The implemented system supports one owner per installation and session-based private uploads. It is not a permanent student information system. Multi-user roles, email recovery, stored interventions and measured improvements in attainment remain outside the current implementation. Its contribution is a reproducible comparison and an accessible staff interface. Any practical educational benefit requires validation in the intended setting.

### 1 4 Development and experimental method

The project combines iterative software engineering with quantitative comparison of prediction procedures. Input validation precedes partitioning. Approximately 60% of records form the fitting set, 20% the calibration set and 20% the test set. Seed 42 fixes the split. Five-fold validation uses only fitting records and identical folds for all candidates. Scaling and encoding are fitted inside each fold [3, 4].

After selection, the predictor is refitted on the fitting partition and held fixed. Only separate calibration residuals determine the interval radius. Test records support error and coverage reporting. Earlier test findings informed this repair, so reuse of these records provides internal benchmarking rather than independent confirmation. No search over seeds or calibration settings was used to improve the test figures.

### 1 5 Acceptance criteria

The system must reject missing G2 and incompatible schemas, preserve outcomes and metadata outside predictors, compare supported candidates consistently, and produce matching displayed and exported estimates. Evaluation must include numerical error, interval coverage and clearly defined alerts. The submission must provide five milestone chapters, presentation material, a demonstration guide and a documented repository. Success does not require a learned model to beat G2; the procedure must allow a strong baseline to win.

# Initial literature review and team roles

### 1 6 Related work

Romero and Ventura [2] review educational data mining and learning analytics as approaches to understanding educational processes from data. This establishes the research context but does not demonstrate that StudentInsight improves attainment. The project therefore evaluates predictive and functional performance, leaving intervention effectiveness to a separate study.

The UCI documentation [1] describes records and questionnaires from two Portuguese secondary schools and identifies the strong relationship between earlier grades and G3. This motivates grade-only comparisons and a stated period-two cutoff. The public files provide historical benchmarks, not proof of compatibility with a local curriculum.

Pedregosa et al. [3] describe scikit-learn's consistent machine-learning API. Its pipelines support repeatable transformations and estimators. Official guidance on leakage [4] explains why preprocessing must learn from training data only. StudentInsight follows that principle within each validation fold and excludes identifiers and G3 from model inputs.

Angelopoulos and Bates [5] introduce conformal prediction for uncertainty sets under exchangeability assumptions. StudentInsight uses split-conformal absolute residuals after fixing the selected predictor. One radius per model yields constant-width intervals. Dataset shift and finite-sample variation constrain interpretation.

Together these sources motivate explicit timing, strong baselines, leakage prevention and transparent uncertainty. The contribution is their integration and verification in a staff workflow, not a new learning algorithm.

| Role | Responsibility | Member |
|---|---|---|
| Coordinator | Scope, schedule, report and demonstration | To complete |
| Data and modelling | Provenance, schemas, comparison and calibration | To complete |
| Application | Authentication, interface and class analytics | To complete |
| Testing and documentation | Checks, evidence, installation and slides | To complete |

These are proposed responsibility areas. Members may combine roles. The group must enter names, student numbers and actual contributions; no individual contribution or supervisory approval is asserted.

# Chapter 2 Design

## 2 1 Architecture

The design separates staff interaction, validation, modelling, persistence and outputs. Streamlit hosts the authenticated workspace. Validation and prediction remain ordinary Python modules, allowing the interface, notebook and tests to call the same implementation. This reduces the risk of different interfaces presenting inconsistent forecasts.

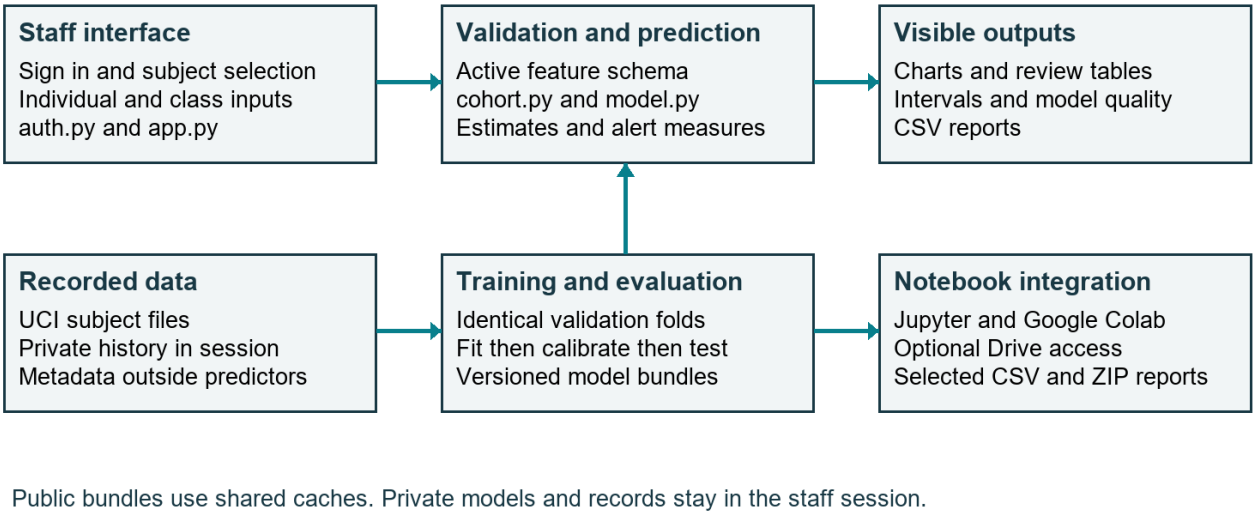



*Figure 1 Implemented components and data flow. Notebook access does not synchronize the Streamlit account database.*

The public route loads or prepares a compatible subject bundle containing a predictor, evaluation metadata and test predictions. The service verifies provenance and integrity before use. Incompatible artifacts trigger an explicit whole-bundle preparation notice; a new predictor is never paired with old intervals. Private history and models remain outside shared public caches.

A staff user signs in, selects a subject and chooses class analysis, individual analysis, model quality or the data guide. Individual inputs follow the active schema. Class uploads retain student codes and class labels as metadata and support filters and downloads. Model-quality results describe historical test records, not the uploaded class.

SQLite stores owner credentials and session information. Datasets and model artifacts are separate filesystem assets. The local owner account does not automatically exist on a hosted copy. The current implementation does not provide multi-user roles or secure permanent student histories.

# Data pipeline and feature schemas

### 2 2 Input and target definitions

G3 is the target and never a predictor. Student identifiers, class names and provenance metadata remain outside fitting. Optional G1 supports the recorded progress indicator G2 minus G1 when a G2-only predictor wins. That difference is derived from the two grades, not independent new information.

| Schema | Model inputs | Purpose |
|---|---|---|
| persistence_v1 | G2 | Persistence and mean candidates |
| grades_v1 | G1 and G2 | Small ridge and forest |
| expanded_v1 | G1, G2, studytime, failures, traveltime, schoolsup, famsup, paid, activities, internet, higher | Expanded comparisons |

Validators require finite whole-number grades from zero to twenty. Study time and travel time use supported ordinal categories; failures use the observed supported range zero to three; support fields accept yes or no. Missing required inputs are rejected. Missing optional G1 remains missing, producing unavailable progress. The mean candidate retains G2 as an interface requirement to preserve the timing contract even though its numerical forecast ignores G2.

The fitting/calibration/test counts are 237/79/79 for Mathematics and 389/130/130 for Portuguese. A common validated record set is partitioned before comparison. Candidate folds, final partitions and dataset hashes are recorded. Public subjects remain separate because some students appear in both subject files.

The public source is UCI Student Performance [1], licensed CC BY 4.0. The downloaded Halabja file contains annual marks on a different scale and lacks matching period definitions; it remains excluded. Attendance and assignments require genuine snapshots available by the cutoff. Future collection should include school, cohort and subject metadata; cutoff and recording timestamps; sessions possible and attended; assignments due, submitted and late; and the later outcome date. Missing denominators must not be replaced with zero.

# Algorithm selection and ethical design

### 2 3 Candidate selection

Six procedures are compared: training mean, G2 persistence, ridge on two grades, ridge on expanded inputs, forest on two grades and forest on expanded inputs. Ridge uses alpha 10. Forests use 250 trees, maximum depth 10, minimum leaf size 3 and seed 42. These modest fixed settings suit the limited dataset. Forests average multiple trees, while ridge penalizes large coefficients in a linear rule [3, 6].

Validation MAE is the primary criterion. Candidates within 0.02 points of the minimum mean validation MAE are effectively tied. Preference is G2 persistence, mean, two-grade ridge, expanded ridge, two-grade forest and expanded forest. This rule precedes test evaluation. Calibration and test errors do not choose the predictor. Fold standard deviation measures variation, not a confidence interval.

### 2 4 Calibration

The quantile rank is found by multiplying the target coverage by one plus the number of calibration residuals, then rounding upward. The radius is the absolute error at that rank in ascending order. A rank exceeding the calibration count is rejected as an unattainable finite quantile. At c = 0.8, the ranks are 64 of 79 and 105 of 130. The predictor remains fixed before calibration, with distinct fitting, calibration and test rows [5].

Raw predictions and intervals remain unchanged through calibration, evaluation, display and export. No display-only clipping occurs. A coverage target is not an individual's passing probability. A grade percentage is only a scale conversion.

### 2 5 Ethical analysis

The workflow uses student codes, session-only uploads, human review and explicit uncertainty. These controls reduce some risks but do not establish fairness, privacy compliance or intervention benefit. Excluding demographics does not remove proxy effects. Staff should avoid permanent labels and record contextual reasons for consequential decisions.

Institutional use requires applicable approvals, an access and retention policy, and subgroup and temporal evaluation. Drive access is optional and requires user authorization. Existing files are not overwritten. A future multi-user deployment needs explicit roles, audit logs, backups and recovery, rather than assuming the prototype provides them.

# Chapter 3 Prototype

### 3 1 Core implementation

model.py implements versioned schemas, estimators, splitting, selection, calibration and numerical reports. model_service.py manages whole-bundle consistency and public-data recovery. subject_training.py validates completed history for additional subjects and retains private models in the signed-in session. The Streamlit layers call these functions rather than reimplementing the mathematics.

Both public predictors currently require G2 only. Optional G1 supports progress reporting. As a behavior example, a G2 value of 12 produces forecast 12 and interval 11 to 13. This is an input-output illustration, not an independently observed outcome or a passing probability.

### 3 2 Training comparison

The comparison table reports validation MAE and fold standard deviation on identical fitting-data folds. G2 persistence has the lowest validation MAE for both subjects and wins without relying on the tie preference. It has no learned predictive weights. Learned candidates remain available for comparison and for appropriate new subject history.

| Candidate | Schema | Math MAE / SD | Portuguese MAE / SD |
|---|---|---|---|
| G2 persistence baseline forecast | persistence_v1 | 0.955 / 0.300 | 0.797 / 0.147 |
| Training-mean baseline forecast | persistence_v1 | 3.540 / 0.057 | 2.502 / 0.150 |
| Ridge: G1 + G2 | grades_v1 | 1.167 / 0.257 | 0.827 / 0.131 |
| Ridge: expanded inputs | expanded_v1 | 1.196 / 0.230 | 0.839 / 0.124 |
| Random forest: grades | grades_v1 | 1.223 / 0.307 | 0.866 / 0.126 |
| Random forest: expanded | expanded_v1 | 1.278 / 0.275 | 0.914 / 0.090 |

The individual report includes predicted G3, percentage equivalent, observed G1-to-G2 change when available, predicted change from G2, reference margin and interval. Class analysis aggregates recorded and predicted grades and displays filtered review rows. Review categories are rule-based indicators, not separately learned labels. The required input schema and model-version context travel with prediction exports.

# Measured results and historical comparison

### 3 3 Reproduced evaluation

The earlier selected procedure was expanded ridge. The audit reproduced saved and retrained predictions to numerical precision and verified data hashes. Original artifact checksums remain unchanged. New results are stored separately so the unfavorable historical report is not erased.

| Measure | Mathematics before to after | Portuguese before to after |
|---|---|---|
| MAE | 1.415 to 1.000 | 0.755 to 0.677 |
| RMSE | 2.195 to 2.072 | 1.159 to 1.170 |
| R2 | 0.765 to 0.791 | 0.862 to 0.860 |
| Interval coverage | 68.4% to 83.5% | 79.2% to 93.8% |
| Average width | 2.896 to 2.000 | 2.013 to 2.000 |

MAE is average absolute prediction error. RMSE gives greater weight to large errors. R2 compares squared error with variation around the test mean; it can be negative and is not an accuracy percentage. MAE decreases for both subjects, but Portuguese RMSE rises slightly and R2 falls. Uniform improvement across all measures would be an incorrect claim.

With nominal coverage 80%, Mathematics covers 66 of 79 test outcomes and Portuguese covers 122 of 130. Calibration counts are 79 and 130, respectively. Both selected predictors have radius one and constant width two points. The interval is not personalized uncertainty.

The original Mathematics undercoverage did not arise from an incorrect conformal order statistic. Its mean absolute calibration error was about 1.136 compared with test MAE about 1.415. The realized test residual distribution was less favorable. This describes the split without establishing a causal explanation. The repair did not widen intervals to fit test coverage; it selected by fitting-data validation and recalibrated on reserved calibration records.

These reused test records are not independent confirmation. Future evaluation should use later cohorts or excluded institutions, following a protocol fixed before observing the new outcomes.

# Alert evaluation and prototype limitations

### 3 4 Below reference alerts

At reference grade 10, a positive outcome means recorded G3 is strictly below 10. An alert means predicted G3 is strictly below the same value. Equality belongs to the at-or-above group. Precision is the share of alerts with a recorded below-reference outcome. Recall is the share of those outcomes correctly flagged. F1 is their harmonic mean. TP, FP, FN and TN denote true positives, false positives, false negatives and true negatives. An undefined denominator displays Unavailable; a defined zero remains zero.

| Subject | TP | FP | FN | TN | Precision | Recall | F1 |
|---|---|---|---|---|---|---|---|
| Mathematics | 26 | 6 | 1 | 46 | 0.813 | 0.963 | 0.881 |
| Portuguese language | 14 | 11 | 1 | 104 | 0.560 | 0.933 | 0.700 |

Mathematics flags 32 of 79 test records, including 26 recorded below-reference outcomes. Portuguese flags 25 of 130, including 14 recorded below-reference outcomes. High recall can therefore coexist with false alerts. These counts describe potential review workload rather than justify automatic adverse decisions. Changing the staff reference grade does not refit the regression predictor or its interval.

The prototype establishes tested prediction, validation, separation and export behavior. It demonstrates that G2 is a strong baseline for these late-course tasks. It does not establish pre-G2 early-warning performance, subgroup reliability, attainment improvements or local institutional generalization.

The records are small, historical and context-specific. Questionnaire timing may differ from prospective availability. Random record splits are not temporal or school-held-out evaluation. Feature importance describes reliance, and changing an input does not prove an intervention effect.

The separate dependency-free application has its own procedure and split generator. Its historical metrics must not be mixed with the repaired Streamlit results. Figures in this chapter describe schema 3 unless explicitly labelled as the reproduced earlier comparison.

# Chapter 4 Integration

### 4 1 Staff workspace and artifacts

The workspace combines sign-in, subject selection, class analytics, individual analysis, model quality and data guidance. Authentication uses salted PBKDF2-HMAC-SHA256 hashes, hashed session tokens, expiry and throttling. Tests exercise setup closure, failed login, sign-out, expiry and revocation. This remains one owner per installation, not multi-user authorization.

Public loading verifies feature and schema versions, data and code fingerprints, library version, split settings, calibration and bundle identity. Manifests detect mismatched file contents. Saved predictions and metrics are checked against the active model. If needed, a new complete bundle is prepared in memory with a visible notice, preserving historical files.

### 4 2 Notebooks and Google Drive

The notebook imports the same modules, compares both public subjects, displays charts and exports reports. Local Jupyter uses the project environment. Colab clones the GitHub repository. Neither automatically synchronizes Streamlit accounts or private session state.

Optional Drive access mounts the user's Drive after Google authorization [8]. The workflow reads a named CSV under StudentInsight/datasets and writes a uniquely named ZIP under StudentInsight/reports. Validation checks required inputs; copied bytes are verified; existing reports are not overwritten. Local Jupyter may use a user-specified synced folder. Actual Google authorization and remote synchronization have not been exercised from a Google account during verification.

### 4 3 Repository and reproducibility

Repository: https://github.com/GamerStudioszw/Student-Performance-Prediction. Application code commit 88b1003 is this report's evidence baseline. README.md explains installation, data, training, tests, notebooks and limitations. Separate requirement files support the application and notebooks. The earlier chapter-one proposal is preserved.

Accounts, private uploads, local outputs and model binaries are excluded from Git. Public audit JSON and evaluation CSV files preserve evidence. New environments can prepare compatible public bundles from the bundled records. Saving a notebook ZIP to Drive does not deploy a model to the institution.

# Testing and operating documentation

### 4 4 Verification

The complete available suite passed 56 tests on 26 October 2026. Checks cover baseline selection, target and metadata exclusion, fold-fitted preprocessing, feature-specific forms and uploads, quantiles, partition separation, threshold equality, undefined scores, export consistency, artifact rejection, authentication and Drive file operations. These tests verify specific behavior, not the absence of all possible defects.

| Test module | Passing tests |
|---|---|
| test_app.py | 12 |
| test_auth.py | 4 |
| test_cohort.py | 7 |
| test_lite.py | 6 |
| test_model.py | 12 |
| test_model_service.py | 6 |
| test_notebook_storage.py | 5 |
| test_subject_training.py | 4 |
| Total | 56 |

The entire notebook also executed with Drive disabled and with a temporary connected folder. Exports matched bundle identities. Synthetic class rows tested software only and did not support predictive-quality claims. Jupyter server startup and authenticated ZIP download passed. Real Google authorization, remote synchronization, a new visual browser review and the assessed live demo remain unverified.

For the staff app, run powershell -ExecutionPolicy Bypass -File .\run.ps1 -Full from the project directory. It installs requirements in .venv and starts Streamlit on port 8502. The equivalent direct launch is .\.venv\Scripts\python.exe -m streamlit run app.py --server.port 8502. Select that interpreter in the IDE; a previous missing-sklearn error resulted from using another Python installation.

Sign in or complete one-time owner setup. Select a subject, review model quality, enter a profile or upload a schema-compatible class CSV, and download reports as needed. Private records and models are lost on sign-out, session loss or restart. Retain original authorized data.

For notebooks, run .\run-notebook.ps1 and choose StudentInsight (.venv). Colab users open the GitHub notebook and run cells in order. Enable USE_GOOGLE_DRIVE in section 1a and authorize access if desired. Additional subjects need genuine completed history; 50 valid distinct records is a technical minimum, not proof of reliability.

# Chapter 5 Final submission

## 5 1 Conclusions

The principal finding is that G2 persistence wins the defined validation comparison for both public subjects. The procedure must permit this result rather than prefer complexity for its own sake. The completed engineering work aligns selected inputs, calibration, displays and exports, preserves historical evidence and communicates prediction timing and limits.

These results do not show improved attainment or institution-wide readiness. Future research requires genuine local subject histories with timestamps, a predefined temporal or school-held-out evaluation, subgroup error analysis and educator review of alert workload. Attendance and assignments should enter training only after cutoff availability is verified. A G1-only early-warning task needs separate training and calibration.

## 5 2 Handover and demonstration

The companion slides follow the five milestone chapters. The separate guide allocates 15 minutes across the problem, design, comparison, staff workflow, exports, notebook integration and limitations. Complete live Google authorization during preparation if showing Drive. Otherwise show the implemented connection cell and explain that cloud verification is pending.

The package prepares the project for assessment. It does not certify supervisor approval or completion of the assessed demonstration. Group members must complete names, numbers and roles, review the evidence and rehearse with an authorized account. Future software work includes explicit multi-user permissions, audited histories, backups, recovery and an institutional deployment review.

| Milestone | Prepared evidence | Remaining activity |
|---|---|---|
| M1 Proposal | Problem, method, literature and roles | Fill names, numbers and allocations |
| M2 Design | Architecture, schemas and ethics | Supervisor review |
| M3 Prototype | Working model comparison and metrics | Independent local validation is future work |
| M4 Integration | App, notebook, Drive support and tests | Google authorization and visual acceptance |
| M5 Final | Report, slides and demo guide | Rehearsal, live delivery and submission |

# References

[1] Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. DOI 10.24432/C5TG7T. https://archive.ics.uci.edu/dataset/320/student+performance

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[7] Project Jupyter. Installing Jupyter. Accessed 26 October 2026. https://jupyter.org/install

[8] Google. Colaboratory frequently asked questions. Accessed 26 October 2026. https://research.google.com/colaboratory/faq.html

[9] StudentInsight repository, code baseline 88b1003. https://github.com/GamerStudioszw/Student-Performance-Prediction

Project-specific evidence is in artifacts/v3/mat/latest.json and artifacts/v3/por/latest.json, which identify immutable evaluation runs. The reproduced historical comparison is in artifacts/audits/legacy-20261022T090956248083Z.json. Run metrics contain fold scores, row partitions, fingerprints and interval settings. Tests are under tests/.

The original proposal remains at documents/Student_Performance_Prediction_Proposal_Chapter_1.docx. Its earlier mathematics-only scope and chapter arrangement are superseded for this submission by the five milestone chapters. No institutional study, participant recruitment or supervisor approval is implied.